隐藏翻译（T）

What are truffles, and why are they so expensive?

【TED科普：白松露为什么能卖出天价？】

Summary: In 2007, a record-breaking white truffle sold for $330,000, highlighting their status as one of the world's most expensive foods. Truffles are rare due to their specific growing requirements, symbiotic relationships with certain trees, and subterranean growth that relies on animals for spore dispersal. Their powerful aroma, crucial for attracting animals and human desire, comes from complex volatile compounds. Despite attempts at cultivation, truffle farming remains highly uncertain because the fungi's reproductive process is not fully understood, and environmental factors are difficult to control. Deforestation and climate change further threaten wild truffle populations, increasing their rarity and cost.

摘要： 2007年，一块破纪录的白松露以33万美元天价售出，彰显了其作为全球最昂贵食材之一的地位。松露的稀缺源于其苛刻的生长条件、与特定树木的共生关系以及依赖动物传播孢子的地下生长特性。其浓郁香气由复杂的挥发性化合物构成，既能吸引动物，也令人类着迷。尽管已有人工培育尝试，但由于松露的繁殖机制尚未完全破解且环境因素难以控制，商业化种植依然充满不确定性。森林砍伐和气候变化进一步威胁野生松露种群，加剧其稀有性和价格攀升。

⏱ Estimated Reading Time: 7 min.

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In autumn 2007, a Tuscan father and son unearthed one of the largest white truffles ever found.

2007年秋天，一对托斯卡纳父子发掘出了有史以来最大的白松露之一。

It later sold at auction for a record-breaking $330,000— more than six times the price of its weight in gold at the time.

后来它在#拍卖会#上以破纪录的33万美元成交——是当时等重黄金价格的六倍多。

Truffles are one of the world’s most expensive foods— in part because global demand often outstrips supply.

松露是世界上最昂贵的食物之一——部分原因是全球需求常常超过供应。

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But why are truffles so rare?

但松露为何如此稀有?

And why don't we just farm more of them?

我们为何不扩大人工种植规模呢?

The answers lie in the truffle’s unique and somewhat mysterious biology.

答案在于松露独特而略带神秘色彩的生物特性。

Truffles refer to nearly 200 species of fungi, of which around 30 are used in commercial trade.

松露包含近200种真菌，其中约30种用于商业贸易。

The most sought-after truffles are native to Europe, confined to regions with lime-rich dry soil and light summer rains.

最受追捧的松露原产于欧洲，仅生长于富含#石灰质#的干燥土壤且夏季雨量适中的区域。

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Each species grows under the canopy of specific trees, where they form tight symbiotic relationships.

每个品种都在特定树种的#树冠#下生长，与树木形成紧密的共生关系。

A truffle’s mycelium, the vegetative root-like structure of the fungus, wraps itself around the tree roots and swaps water and nutrients for sugars.

松露的菌丝体——即#真菌#的须根状营养结构——会缠绕树根，用水分和养分交换糖类。

While truffles themselves aren’t mushrooms, the parts of truffles we covet are, like mushrooms, the fruiting body of the fungus.

虽然松露本身不属于蘑菇，但我们#垂涎#的松露部位与蘑菇一样，都是#真菌#的子实体。

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But while mushrooms grow above ground, allowing their spores to be dispersed by both wind and foraging animals, truffles grow entirely below the soil.

但蘑菇生长在地面上，孢子能通过风和觅食动物传播；而松露完全在地下生长。

So, they release a powerful aroma to lure hungry forest animals to come to do their spreading.

因此它们会释放强烈#香气#来#引诱#饥饿的森林动物帮助传播孢子。

If these animals happen to drop these spores near the roots of a particular type of tree, only then might a truffle develop, several years later.

只有当这些动物恰好将孢子遗落在特定树种根部附近时，数年后才可能长出松露。

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This pungent aroma is also what’s made truffles alluring to humans.

正是这种#浓烈#的#香气#让松露对人类充满诱惑。

In fact, much of what we perceive as truffle flavor comes from its unique scent.

事实上，我们#感知#到的松露#风味#主要来自其独特的#气味#。

A single species typically emits 30 to 60 different volatile compounds, which are produced by both the truffle and the bacteria and yeast that colonize them.

单个松露品种通常释放30至60种挥发性化合物，这些物质由松露及其附生的细菌和酵母共同产生。

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But despite their scent, finding these subterranean gems within a dense forest is far from easy.

但尽管有#气味#指引，在#茂密#森林中寻找这些地下珍宝仍非易事。

As early as the 1400s, Europeans used pigs as truffle-finding companions.

早在15世纪，欧洲人就开始用猪作为寻找松露的伙伴。

Truffles emit dimethyl sulfide, a compound that reeks of cabbage, which helps draw the pigs in.

松露会#释放#带有#卷心菜#气味的二甲硫#化合物#，这对猪极具吸引力。

But when a pig finds a truffle, they're apt to take a bite out of the profit.

但猪发现松露时，#容易#咬上一口导致利润受损。

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So many truffle hunters began replacing pigs with dogs who, while not natural-born truffle hunters, are more easily trained.

于是许多猎松人开始用狗替代猪——狗虽非天生会找松露，但更易训练。

Meanwhile, others learn to track their prize by following species of flies that are known to lay their eggs near truffles.

与此同时，有人通过跟踪常在松露附近产卵的蝇类来定位目标。

Nevertheless, today, the global market still demands more truffles than any pig, dog, or fly can sniff out.

然而如今，全球市场的松露需求量仍远超猪、狗或苍蝇能#嗅出#的数量。

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And truffles continue to grow even rarer— and more expensive— as deforestation and climate change shrink their suitable terrain.

随着#森林砍伐#和气候变化不断#压缩#适宜生长的#地域#，松露正变得愈发稀有和昂贵。

So why don't we just farm them?

那我们为何不进行人工种植呢?

We do, to some extent.

其实我们已在尝试。

In 1808, a French farmer became the first to cultivate truffles when he planted acorns under truffle-bearing oaks, and then transplanted the seedlings with— unknowingly at the time— some fungi companions.

1808年，一位法国农民将橡树下采集的橡子连土移植，无意中携带了真菌伙伴，成为人工培育松露的第一人。

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By the early 1970s, scientists cracked how to artificially inoculate tree seedlings, which allowed growers to sow certain species of truffles in orchards far outside their native range.

到20世纪70年代初，科学家攻克了树苗人工接种技术，使种植者能在原产地以外的果园培育特定松露品种。

However, truffles are still not considered a domesticated crop.

但松露仍未被视为驯化作物。

There are just too many unknowns, as researchers are still working out the precise temperature, precipitation, elevation, soil pH, and other environmental factors that determine a truffle orchard's success.

未知因素太多，研究人员仍在破解决定松露果园成败的温度、降水、海拔、土壤pH值等精确环境参数。

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Part of the problem is that truffles need to mate to form that valuable fruiting body, yet truffles sex is still largely a mystery.

部分难题在于：松露需要交配才能形成珍贵的子实体，但其性别机制仍是个谜。

Researchers know that these fungi are hermaphrodites that cannot reproduce alone.

研究者已知这些真菌是雌雄同体但无法独自繁殖。

So, when a truffle reproduces, one fungus takes on the maternal role and the other the paternal role, each contributing half of the genes that will form the spores.

当松露繁殖时，一个菌株承担母本角色，另一个承担父本角色，各自为孢子形成提供一半基因。

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However, when researchers search the area near a truffle fruiting body, they only ever find maternal mycelium.

然而研究人员在松露子实体附近只能找到母本菌丝体。

Where the paternal partner lives— or hides— remains unknown.

父本伴侣存在于何处——或藏身何处——仍属未知。

So each time a hopeful truffle enthusiast begins a new farm, there's no guarantee truffles will grow, and they'll have to wait four years or more to find out.

因此每当满怀希望的松露爱好者开辟新种植园时，无法保证一定能长出松露，且需等待四年或更久才能知晓结果。

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But many take the gamble, as growers know that if they're patient and lucky, they might just dig up gold.

但许多人仍愿冒险一搏，因为种植者明白：若有耐心且运气够好，他们或许真能挖到"黄金"。